

Indian Institute of Technology (IIT-Bombay)

AUTUMN Semester, 2026

COMPUTER SCIENCE AND ENGINEERING

CS230: Digital Logic Design and Computer Architecture

Tutorial - I

Full Marks: 0

Time allowed: ∞ hours

1. Simplify the following Boolean function using a Karnaugh Map and express the final result in the minimal Sum of Products (SOP) form:

$$F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$$

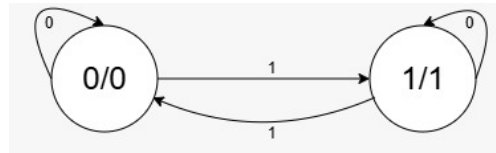
2. Design a synchronous sequential circuit using D flip-flops that acts as a one-input/one-output sequence detector. The circuit should produce an output value of 1 every time the overlapping sequence **110** is detected, and 0 otherwise. Provide the state transition table and derive the minimal Sum of Products (SOP) Boolean expressions for the D flip-flop inputs and the final output.
3. The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ')$, written F^D , is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. What are the total number of self-dual functions with n Boolean variables?
4. Consider a circuit realization of a combinational logic block as shown in figure to obtain the following truth table:

A	B	Q_{n+1}
0	0	$\overline{Q_n}$
0	1	1
1	0	Q_n
1	1	0

The combinational logic involves:

- a) Only Ex-OR gates
- b) AND and NOT gates
- c) NOT and Ex-NOR gates
- d) OR and NAND gates

5. Consider the state diagram of the Finite State Machine (FSM) provided below:

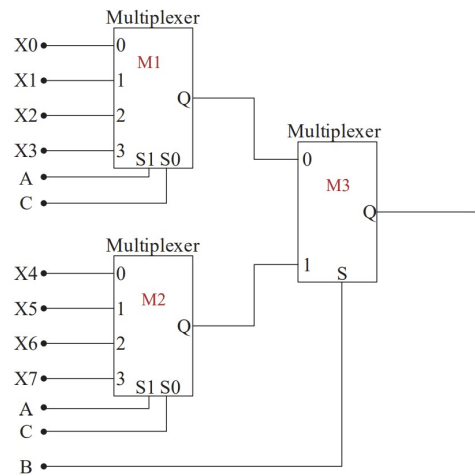


- Write the output sequence for the input sequence: 001010110110111
- What is the functionality of the above FSM?
- Which flip-flop can be used to implement the above FSM without any external logical gates?

6. Describe a finite state machine that will detect three consecutive coin-tosses (of one coin) that result in heads.

7. What values of $(X_0, X_1, X_2, X_3, X_4, X_5, X_6, X_7)$ will realize the Boolean function:

$$\overline{A} + \overline{A} \cdot \overline{C} + A \cdot \overline{B} \cdot C ?$$



8. Design a combinational circuit to realize the following function, where $X = x_2x_1x_0$ is a 3-bit binary input and $F(X) = F_2F_1F_0$ is the 3-bit binary output:

$$F(X) = \begin{cases} X + 1, & 0 \leq X \leq 3, \\ X - 1, & 4 \leq X \leq 7. \end{cases}$$

Determine the minimal SOP and POS.

9. Design a synchronous sequential circuit using two flip-flops Q_1 and Q_0 , and one control input x . The state is represented as Q_1Q_0 , where Q_1 is the **MSB** (leftmost bit) and Q_0 is the **LSB** (rightmost bit).

For $x = 0$, the required sequence is

$$00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00 \rightarrow \dots$$

and for $x = 1$,

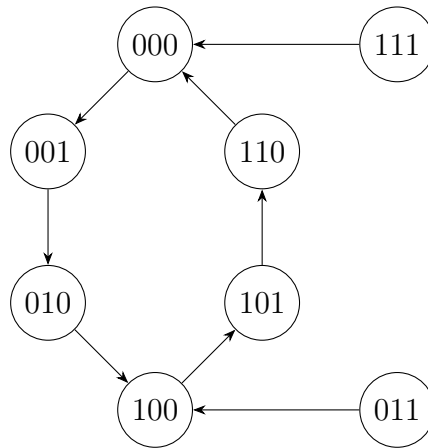
$$00 \rightarrow 11 \rightarrow 10 \rightarrow 01 \rightarrow 00 \rightarrow \dots$$

Determine the required flip-flop inputs for:

- (a) D flip-flop realization: D_1, D_0
- (b) T flip-flop realization: T_1, T_0

10. Design a synchronous counter that follows the state transitions specified by the given state diagram, using D flip-flops.

Given State Diagram



11. Realize the following Boolean function using one 4 : 1 multiplexer and at most one additional gate.

$$F(A, B, C) = \Sigma m(1, 2, 5, 6)$$

Use A and B as the select inputs of the multiplexer.

Determine the required MUX data inputs I_0, I_1, I_2, I_3 and draw the resulting circuit.
